

unicast transmission (see Box: Unicast and multicast). However, combining D2M broadcast with unicast can create a hybrid and integrated network, enhancing user

Unicast and Multicast

Unicast and multicast are two different techniques for data transmission in 5G broadcasting. Unicast, the most common technique used for data transmission, is a one-to-one communication to send data from one device to another and is used for streaming video, downloading files, and making phone calls. Multicast provides a one-to-many communication to send data transmission from one device to a group of devices and is used for broadcasting live TV, delivering updates to a fleet of devices and sending emergency alerts etc.

experience. The converged network leverages the best of unicast and multicast techniques to enable innovative applications such as free-to-air media content delivery, robust emergency notifications for public safety, and enhanced venue casting.

Free-to-air media content delivery

A soccer match livestreamed over unicast cellular connections and watched simultaneously by a large audience creates immense strain on the unicast network, reducing service quality and increasing delivery costs. The red curve in Fig. 2 illustrates

data traffic peaking when accessed by many consumers using the unicast network. D2M broadcast dynamically delivers the same content, significantly reducing peak traffic on the unicast network (point #1 in Fig. 2).

This approach ensures robust, uninterrupted video playback while freeing unicast network resources for other content access. High-mobility users moving outside the D2M broadcast service area (point #2 in Fig. 2) continue accessing the same content seamlessly by switching to the unicast network. When the system detects the same content on the broadcast network, it automatically switches to it. Similarly, it reverts to the unicast network without interruption when D2M coverage ceases. This automatic, seamless switching extends the effective service area. Additionally, content consumed through the D2M network does not count towards monthly data caps (Fig. 3).

Delivering emergency notifications during disaster events

During disasters like floods, earthquakes, or wildfires, cellular networks often fail due to structural damage. However, D2M broadcast infrastructure (Fig. 4) can deliver emergency alerts directly to mobile devices without requiring internet connections. Recently, Sichuan Radio and Television in China used D2M to transmit emergency alert signals generated in Chengdu. The alerts were instantly displayed on Qualcomm-powered test devices in both active use and standby modes. This demonstration highlights how public safety agencies can utilise D2M for alert dissemination during emergencies.

Enhanced venue casting at events

Live sports events can leverage D2M broadcast for venue casting without overloading cellular networks. Recently, Austria's ORS, a broadcasting service provider, used a D2M broadcast van equipped with an encoder, modulator, and transmitter to deliver four high-quality live video feeds from cameras around a racetrack during a MotoGP event. Attendees enjoyed uninterrupted, low-latency live feeds on their smartphones, demonstrating D2M's capability for live video delivery without impacting nearby cellular towers.

Challenges in implementation

Large-scale deployment of D2M technology in India faces challenges such as adapting existing infrastructure, which requires an estimated investment of ₹200-300 billion. Spectrum availability remains problematic due to a congested radio frequency landscape. Agreements and licensing protocols between broadcasters, mobile operators, and content creators must also be finalised.

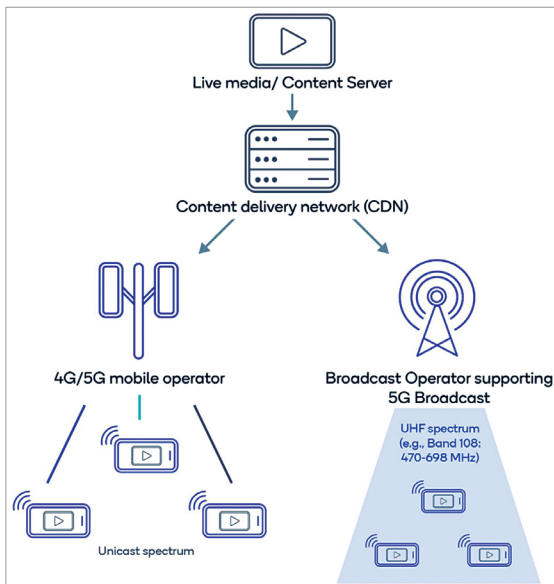


Fig. 3: Extending the effective service area by combining unicast and D2M broadcast (Source: IIT Kanpur, June 2022 white paper)



Sports events can be venue casted using D2M broadcast vans (Source: <https://www.live-production.tv>)